When Structure Hurts: Understanding Evidence Loss in Retrieval-Augmented Reasoning

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Abstract

Retrieval-augmented systems increasingly turn retrieved passages into sentence lists, reasoning orders and relation graphs before generation. System-level comparisons cannot tell which part of such a transformation helps or hurts, because retrieval, content and formatting change together. We hold the evidence fixed: gold supporting passages for multi-hop questions are presented in representations whose fact sets are verified identical per question, and we vary segmentation, order, relation lines and their labels one at a time. In a pre-specified test on 4,976 MuSiQue and 2WikiMultiHopQA questions with Qwen2.5-7B-Instruct, graph-derived ordering and derived relation lines had no detectable effect, and splitting passages into sentence items lowered 2WikiMultiHopQA exact match by 1.7 points, through abstention; the latter did not replicate with Mistral-7B. Exploratory fixed-order interventions then located the cost of predicted structure in a single label: showing the `CONTRADICTS` relations predicted by an NLI classifier cost about 3 points, and relabelling only those lines recovered it. A second pre-specified test on 6,400 untouched questions confirmed the effect (2WikiMultiHopQA -7.7 points, HotpotQA -5.3 points among affected questions), mostly through abstention and partly through wrong answers; it also held for Mistral. Whether the harmful contradictions are false is supported only by automatic triage, and all results concern 7B readers.

1 Introduction

Retrieval-augmented generation (RAG) gives a language model access to text that its parameters do not contain, by retrieving passages and placing them in the prompt (Lewis et al., 2020). For multi-hop questions, as in MuSiQue, 2WikiMultiHopQA and HotpotQA (Trivedi et al., 2022; Ho et al., 2020; Yang et al., 2018), the passages are usually handed to the model as a list, and finding and linking the relevant sentences is left to the model.

A growing family of methods therefore transforms retrieved evidence before generation. Passages are decomposed into propositions or claims (Chen et al., 2024), organised into trees of summaries (Sarthi et al., 2024), or linked into graphs whose nodes, relations and paths are serialised into the prompt (Edge et al., 2024; Gutiérrez et al., 2024, 2025; Zhu et al., 2025; Guo et al., 2024; Chen et al., 2026). Each such transformation can also go wrong. Splitting a passage into sentences turns "It stars Anne Hathaway ..." into an item whose subject must be recovered from elsewhere; a token budget may cut the sentence that names the answer because the format spends tokens on identifiers and relation lines; and a relation predictor may assert a relation the sources never made. In our experiments an NLI classifier marked "The teaser of the film was released in February 2016" and "The trailer of the film was released in October 2016" as contradicting each other, and, shown that line, the model declared the evidence insufficient for a question it otherwise answered correctly.

These costs are hard to separate from the benefits in existing evaluations. When a graph-based system is compared with passage RAG, its retrieval, the content that reaches the reader and the way that content is formatted all differ at once (Xiang et al., 2026; Han et al., 2025; Zhou et al., 2025). The modular analysis of Nishida et al. (2026) shows that noisy extracted triples hurt GraphRAG. There too, however, extraction errors act through what is retrieved, so the effect of an incorrect relation *as presented to the reader* remains unmeasured.

We argue that the effect of an evidence representation should be measured with the evidence held fixed, and that each component of a structured rep-

resentation should be varied on its own: sentence segmentation, order, relation lines, and their provenance and labels. We therefore ask: when the same supporting facts are presented as passages, sentence items, ordered items or items with explicit relations, does answer accuracy change, and which component causes the change?

We present the gold evidence of each question in several representations under two protocols. An *information-controlled* protocol verifies per question that every representation shows the same facts; a *budget-controlled* protocol fixes a representation-independent fact ranking and token budget and records which facts each format can afford. After an exploratory pilot, we tested six contrasts, fixed before any generation, on a disjoint sample of 4,976 questions with Qwen2.5-7B-Instruct.

That test was largely null for structure: graph-derived order and derived relation lines did not change exact match detectably, and the one confirmed effect, a small cost of sentence items, ran through abstention and did not recur with a second reader. We then intervened on the predicted relation block one factor at a time. With facts and order fixed, the cost of predicted structure sat in one label: showing `CONTRADICTS` lines made the reader abstain, and relabelling only those lines removed the cost, whereas their number, length and order did not matter. We froze a second protocol for this contrast and tested it on 6,400 untouched questions, including a dataset that had no part in forming the hypothesis; it was confirmed on both, and it held for a second model family.

Our contributions are:

- a controlled framework that separates the facts a representation shows from its segmentation, order, relation lines, relation labels and formatting overhead, with per-question verification of fact identity;
- two pre-specified confirmatory tests: a mostly null test of order, relation lines and segmentation on 4,976 questions, and a confirmed test that asserted contradiction lines lower accuracy on 6,400 untouched questions;
- intervention evidence that the label, not the line count, length or order, carries the cost, that abstention is its main but not only channel, and that it holds across two readers and with real retrieval;
- a reproducible release of code, frozen question sets, run manifests, analysis outputs and blinded annotation packages, with an audit mapping every number in this paper to its source.

2 Related Work

Passages, finer units and compression. Retrieval-augmented generation conditions a language model on retrieved text (Lewis et al., 2020), usually by concatenating passages in rank order. This presentation is not neutral: models use information in the middle of a long context less reliably than at its edges (Liu et al., 2024), and irrelevant passages change generation quality in ways that retrieval scores do not predict (Cuconasu et al., 2024). Dense X Retrieval indexes propositions and finds that finer units help retrieval and downstream QA under a fixed computation budget (Chen et al., 2024). Compression methods trade information for length explicitly (Jiang et al., 2024; Jeong et al., 2025); our budget protocol instead fixes the fact ranking, so that formatting cost is separated from content choice.

Graph-structured retrieval. GraphRAG builds an entity graph with community summaries (Edge et al., 2024), RAPTOR a tree of recursive summaries (Sarthi et al., 2024), and HippoRAG and HippoRAG 2 a knowledge graph traversed with personalised PageRank (Gutiérrez et al., 2024, 2025). KG^{2}RAG links chunks through fact-level relations (Zhu et al., 2025), LightRAG combines graph indexing with dual-level retrieval (Guo et al., 2024), and PathRAG retrieves, prunes and serialises relational paths (Chen et al., 2026). LinearRAG avoids relation extraction altogether, arguing that extracted relations are unreliable and costly (Zhuang et al., 2026). In all of these systems graph construction, retrieval and prompt formatting change together.

Analyses of when structure helps. The closest work to ours is the modular analysis of GraphRAG by Nishida et al. (2026). They vary triple extraction, community clustering and report generation on two templated factoid benchmarks with a GPT-4o reader, and find that the precision and coverage of extracted triples limit downstream accuracy, that noisy LLM-extracted relations hurt both evidence and answer recall, and that simple template

reports outperform LLM summaries. GraphRAG-Bench evaluates the whole pipeline on tasks of increasing complexity and identifies when graphs give measurable benefits (Xiang et al., 2026). A benchmark comparison of RAG and GraphRAG finds task-dependent strengths (Han et al., 2025), and a unified framework compares graph-based RAG methods under the same settings (Zhou et al., 2025). Dai et al. (2024) present the same knowledge-graph facts as linearized triples or as fluent text and find that the format matters and that model preferences differ.

These studies establish that structure does not help uniformly and that extraction errors propagate. In each of them, however, changing the structure also changes what the reader sees, because extraction, clustering and verbalisation determine the retrieved content, or, in the format comparison, because the units themselves change. Our study is narrower. We fix the sentences the reader sees, verify their identity per question, and vary one presentational factor at a time (segmentation, order, relation lines, their provenance and their labels); we also account for formatting tokens under a budget and separate abstention from wrong answers. We do not measure whether structure improves retrieval, which these system-level studies address.

Datasets and attribution. We evaluate on MuSiQue (Trivedi et al., 2022) and 2WikiMultiHopQA (Ho et al., 2020), with HotpotQA (Yang et al., 2018), HoVer (Jiang et al., 2020) and FEVER (Thorne et al., 2018) as secondary pilot datasets. ALCE evaluates citations (Gao et al., 2023); we record citations but do not analyse them. Experiment 0 includes our reconstruction of the EG-RAG evidence-graph pipeline only as a retrieved-evidence baseline. Its unspecified components are our own choices (Appendix C), and no result of the original architecture paper is reused.

3 Problem Formulation

Structured pipelines change *which facts* reach the answerer and *how* they are arranged and annotated; we measure the two separately.

Facts and representations. For a question q with gold answers A_q we start from a fixed set of source passages; in our controlled experiments these are the dataset's gold supporting passages. Segmenting them into sentences gives the fact set $F_q = \{f_1, \ldots, f_n\}$, where each fact keeps the identifier of its source span. Because facts are sentences of the gold passages, not model-generated paraphrases, F_q contains no extraction error. A sentence is not an atomic proposition, however: some sentences need their passage to be read (*It stars* ...), so segmentation can make a fact harder to interpret without removing it. Model-based proposition extraction enters only through the retrieved-evidence systems of Experiment 0, and we never mix extracted claims into an information-controlled comparison. A representation ρ maps the facts, optionally with structure S_q derived from annotations or predicted by a model, to an evidence block

$$E_q^\rho = \mathrm{render}_\rho\big(\pi_\rho(F_q),\, S_q\big), \qquad (1)$$

where π_ρ orders and groups the facts and render_ρ serialises them, possibly with a list of relations. We write $\mathrm{facts}(E)$ for the facts that appear in a block, and $\mathrm{added}(\rho)$ for the information that the block carries beyond them. We distinguish three kinds of added information that structured pipelines usually change together: the *order* π_ρ of the items; explicit *relation lines* $L_\rho \subseteq F_q \times F_q \times \mathcal{T}$, each asserting a typed relation (dependency, support, contradiction, duplication, supersession) between two items; and the *provenance* of those relations, derived deterministically from dataset annotations or predicted by a model. Our controls change one of these at a time, and include a neutral label that asserts no relation type at the same pairs, and a non-relational filler at the same line count. A fixed answerer M with one prompt template τ, identical for every representation, produces $\hat{a} = M(\tau(q, E_q^\rho))$ with greedy decoding. The template allows the answer *insufficient evidence*, so each outcome is one of three states: correct (C), answered but wrong (W), or not answered (A, an abstention or an unparseable reply). Exact match counts A as incorrect; we report the transitions between states separately, because a representation can change accuracy by changing the willingness to answer.

Protocol A: information-controlled. Protocol A compares representations whose fact sets are identical, $\mathrm{facts}(E_q^\rho) = F_q$ for every ρ, which we verify per question by comparing identifiers and text digests. An accuracy difference then cannot come from a missing fact, only from $\mathrm{added}(\rho)$ or formatting tokens. We record the content tokens $c_q = \sum_j |\mathrm{text}_j|$ and the overhead $o_q^\rho = |E_q^\rho| - c_q$

separately.

Protocol B: budget-controlled. Under a budget $B_q = \beta\, c_q$ we rank the facts once, with BM25 against the question, independently of the representation and without gold labels, and each condition keeps the longest prefix of the ranking whose rendering fits,

$$k_\rho = \max\{k : \ |\mathrm{render}_\rho(\pi_\rho(F_q^{\le k}), S_q)| \le B_q\}. \quad (2)$$

Under the *per-representation* policy each representation keeps its own k_ρ, so verbose formats keep fewer facts; under the *shared* policy all keep $\min_\rho k_\rho$ and show the same facts. Facts are never cut mid-sentence. For every dropped fact we record whether it is a gold supporting fact and whether it contains the answer string; this is our measure of information loss.

Objective. For a contrast between ρ and ρ' we estimate the paired difference in question-level exact match over the questions Q available in both conditions, with unanswered questions scored as incorrect:

$$\Delta_{\rho'-\rho} = \tfrac{1}{|Q|} \textstyle\sum_{q\in Q} \big(\mathrm{EM}(q,\rho') - \mathrm{EM}(q,\rho)\big). \quad (3)$$

4 Experimental Framework

Figure 1 (Appendix A) shows the pipeline in the order in which a question passes through it.

Questions and structure. Questions come from frozen samples of the development splits, drawn with a fixed seed before any generation and stored with the hashes of the dataset files; the pipeline refuses to run on any question outside a frozen file or on the questions used during prompt development. The MuSiQue decomposition and the 2WikiMultiHopQA evidence triples give an order of reasoning steps, from which we *derive* dependencies between facts by rule. Where a dependency can be anchored to a fact by an exact string match we call it *exact-anchor derived*, otherwise *fallback-anchor derived*. No edge in this study is human-verified.

Retrieved-evidence baselines (Experiment 0, pilot). Five systems answer from the top four passages of each question's pool: BM25 (B0), reranked (B1), claims extracted from them (B2), a predicted reasoning path (B3), and our reconstruction of the EG-RAG pipeline (B4; Appendix C). Their budgets differ, so we report them as systems.

ID	Rendering of the same fact set
R0	one item per gold passage, dataset order
R1	one item per sentence, dataset order
R2	R1 shuffled (3 seeds, averaged)
R3	R1 grouped by reasoning step
R5	R1 in graph-derived (topological) order
R4	R5 + derived relation lines
R4P	predicted relation lines and order
Fixed-order controls (exploratory)	
R4P_FO	R5 order + predicted relation lines
R4P_FO_NC	R4P_FO without `CONTRADICTS` lines
R4P_FO_CN	R4P_FO, `CONTRADICTS`→`RELATED_TO`
R4P_FO_PAD	R4P_FO, `CONTRADICTS` lines→filler
R4N_FO	R4P_FO, every label `RELATED_TO`
R5P	predicted order, no relation lines
R4P_NC	R4P without `CONTRADICTS` lines

Table 1: Representations. Every item carries a per-prompt identifier (`[E1]`, ...) and its source title.

Representations (Experiments 1 and 2). Table 1 defines the representations; Appendix B shows examples. R5 orders the sentence items by a topological order of the derived dependency graph at passage level, and R4 adds a `Relations:` block listing the derived dependencies. R5−R1 therefore measures the effect of the graph-derived order without relation lines, and R4−R5 the effect of explicit relations at a fixed order. R4P renders the same facts with *predicted* relations: typed relations (support, contradiction, duplication, supersession) from an NLI cross-encoder over candidate pairs, plus a title-mention dependency heuristic, with the order recomputed from the predicted graph.

Selection, generation and evaluation. Under Protocol A every representation shows the complete fact set; under Protocol B the budget is $\beta \in \{1.0, 0.5\}$ times the content tokens. A budget of $1.0\times$ the content tokens is already smaller than the full rendering of any representation that spends tokens on identifiers, titles or relation lines. The prompt (Appendix A) is identical for every representation, asks for an answer and the identifiers of the items used, and allows the answer *insufficient evidence*. A guard checks for every record that the prompt equals the template filled with the evidence block and the question, so no gold answer can enter the prompt through another channel. Exact match (EM) after SQuAD normalisation (Rajpurkar et al., 2016), maximised over aliases, is the primary metric for QA, and verdict accuracy for HoVer and FEVER. Secondary metrics are token F1, coverage (the fraction of

questions with a parsed, non-abstaining answer), answer-in-evidence, and evidence and overhead tokens. Every attempted question stays in the denominator.

Fixed-order relation controls (added after the confirmatory test). R4P differs from R4 in three ways at once: its relation lines are predicted rather than derived, there are more of them, and its item order is recomputed from the predicted graph. To separate these factors we built further representations of the same facts (Table 1, lower part). R4P_FO shows the predicted relation lines at the derived order of R5 and R4. Against it, R4P_FO_NC removes only the predicted `CONTRADICTS` lines; R4P_FO_CN keeps every line, pair and position and changes only the word `CONTRADICTS` to the neutral `RELATED_TO`; and R4P_FO_PAD replaces each contradiction line, in place, by the non-relational line *(this line is intentionally left blank)*, which keeps the number of lines and almost their length (7 against 8–9 tokens). R4N_FO relabels every predicted line as `RELATED_TO`; R5P uses the predicted order without lines; and R4P_NC is R4P without its contradiction lines. Each pairwise contrast changes one factor on identical facts and an identical prompt template; where a block has no contradiction line, the prompts of R4P_FO, R4P_FO_NC, R4P_FO_CN and R4P_FO_PAD are byte-identical. Because these controls were designed after the confirmatory result, we treat them as exploratory interventions, and we then tested the central one prospectively (§5).

Further exploratory conditions. For retrieved evidence, a shared, gold-free stage (BM25 over the question's pool, cross-encoder reranking, top four passages) fixes the passages before any representation is built; gold labels are used only afterwards, to stratify by retrieval outcome. Because the answer instruction offers *insufficient evidence*, we also vary the abstention policy as a separate factor: the frozen prompt, the same prompt without that sentence, and the same prompt with stricter guidance (*If the evidence does not fully establish the answer, reply "Answer: insufficient evidence". Do not guess.*). We also repeat the main contrasts with a second reader.

Perturbations (Experiment 2, pilot). Interventions on the R1 and R4 renderings remove required facts, retype relations, permute items, insert non-gold distractor sentences, strip titles, or corrupt a fact (Appendix D). Experiment 3 (component attribution) requires human-verified structure and was not run, and we did not evaluate an adaptive representation selector.

5 Experimental Setup

Questions. The exploratory pilot (seed 1, fixed before any generation) has 100 MuSiQue and 100 2WikiMultiHopQA questions, plus 50 each from HotpotQA, HoVer and FEVER; the fifteen questions used in prompt development are excluded everywhere. After the pilot we wrote a confirmatory protocol and committed it to version control, and only then drew a *confirmatory sample* with a different seed (2) from the questions that the pilot and prompt development had not used: 1,744 MuSiQue questions (hop 2/3/4: 687/686/371, the last stratum exhausting the eligible pool) and 3,232 2WikiMultiHopQA questions (808 per type). Its size is the number of questions needed for 80% power to detect a difference of 0.03 exact match at $\alpha = 0.05/6$, computed from the pilot standard deviation of paired differences inflated by 20%, maximised over the three contrasts of each dataset. The confirmatory run uses Protocol A only. The protocols were committed to our repository with timestamps before the samples they govern were drawn; they were not lodged with an external registry, so we call them *pre-specified*.

Exploratory and second confirmatory samples. The controls, the second reader and the abstention-policy variants (on 1,000 random confirmatory 2WikiMultiHopQA questions) reuse the confirmatory questions; the retrieved-evidence experiment uses 500 fresh MuSiQue and 500 2WikiMultiHopQA questions (seed 3). Finally, after the controls had located the effect, we wrote a second confirmatory protocol for its central contrast, R4P_FO against R4P_FO_NC, committed and published it, and only then drew *confirmatory sample 2* (seed 5) from questions no earlier sample had used: 2,600 2WikiMultiHopQA questions and 3,800 HotpotQA questions, a dataset that played no role in forming the hypothesis (MuSiQue's eligible pool was exhausted). The protocol fixes the population (questions whose R4P_FO block shows at least one contradiction line, an input property), the test (paired bootstrap, Holm over the two datasets) and a sample size powered for

half the exploratory effect.

Answerer. The answerer is Qwen2.5-7B-Instruct (Qwen Team, 2024) at the pinned revision `a09a3545`, in bfloat16, with greedy decoding, at most 64 new tokens and at most 4,096 input tokens. Auxiliary models are bge-small-en-v1.5 (dense retrieval in B4), ms-marco-MiniLM-L6-v2 (reranking), nli-deberta-v3-xsmall (relation prediction) and nli-deberta-v3-small (grounding), all pinned (He et al., 2023). Experiment 0 also uses the 7B model as its claim extractor. The second reader is Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) at revision `c170c708`, with the same prompt, decoding and evidence; its chat template folds the system message into the user turn. Because a smoke test showed that it often answers in full sentences, we fixed before its run a secondary *lenient* metric (a gold alias contained in the answer), applied identically to both readers.

Hardware and batch size. All generations run on a single NVIDIA RTX 5000 Ada GPU (32 GB; CUDA 13.0, PyTorch 2.12.1, Transformers 5.12.1) with batch size 1; all runs together used about 20 GPU-hours. In a preflight on questions excluded from all reported runs, batched bfloat16 greedy decoding changed 3 of 8 outputs of the 7B model, including one answer, whereas unbatched outputs were identical across runs.

Statistics. The unit of analysis is the question. For each contrast we report the paired difference in question-level EM over the questions available in both conditions, with a 95% percentile bootstrap interval (2,000 resamples, seed 0) and a two-sided bootstrap p-value (Efron and Tibshirani, 1993); R2 seeds are averaged within a question before pairing. Every contrast also reports the number of byte-identical prompt pairs and the estimate on the remaining *informative* pairs. Protocols, datasets, budgets, policies and edge strata are never pooled. All these procedures, the primary metric and the primary contrasts were fixed in a pilot protocol before any generation. The confirmatory family consists of the six primary tests (three contrasts on two datasets) on the confirmatory sample, with Holm correction (Holm, 1979) at a family-wise α of 0.05; the protocol assesses equivalence to zero by whether the 90% bootstrap interval lies inside a margin of ± 0.02 fixed before the confirmatory sample (a per-test TOST). We additionally apply a paired TOST with Holm correction over the six tests; the two procedures agree for every contrast. The second confirmatory family consists of two tests (one per dataset), Holm-corrected, with abstention as a separately corrected secondary outcome. Everything else, including all pilot results, the secondary contrasts on the confirmatory sample, and the analyses that we added after seeing the confirmatory result, is exploratory and reported as such. An interval that contains zero is reported as inconclusive, never as evidence of equivalence.

6 Results

We report the first confirmatory test, then the exploratory interventions it motivated, then the second confirmatory test of the central intervention on untouched questions, and finally the exploratory checks of reader, retrieval and abstention policy. Intervals are paired 95% bootstrap intervals over questions, and every difference is the first-named condition minus the second.

6.1 Confirmatory test 1: segmentation, order and derived relations

Table 2 reports the six pre-specified tests on 1,744 MuSiQue and 3,232 2WikiMultiHopQA questions; all 44,784 records were generated without errors, and the fact sets were verified identical across representations for every question. One test rejects its null hypothesis after Holm correction: on 2WikiMultiHopQA, presenting the gold passages as sentence items (R1) rather than passages (R0) lowers exact match by 0.017 ($[-0.029, -0.005]$, $p_{\text{Holm}} = 0.018$). On MuSiQue the same contrast is -0.001 and equivalent to zero within ± 0.02. Neither structural contrast is significant: the graph-derived order (R5$-$R1) gives -0.011 and -0.008, and derived relation lines at a fixed order (R4$-$R5) give $+0.010$ and $+0.006$ (MuSiQue, 2WikiMultiHopQA); on 2WikiMultiHopQA both are equivalent to zero within ± 0.02. Many pairs are byte-identical (R5 = R1 for 1,858 2WikiMultiHopQA questions), and the structure is derived, not verified; neither exact-anchor nor fallback-anchor edges show an effect (Appendix Table 18).

The confirmed effect runs through abstention. Classifying each outcome as correct (C), answered but wrong (W) or not answered (A), the sentence format raises abstention on 2WikiMultiHopQA by 5.2 points $[4.1, 6.3]$, while accuracy among questions answered under both formats is unchanged

Data	Contrast	n	ΔEM	95% CI	p	p_{Holm}	+/−	Ident.	Inform. Δ	ΔCov.	Equiv.
MuSiQue	R1−R0	1744	-0.001	[-0.014, +0.014]	0.974	0.974	70/71	3	-0.001	+0.009	yes
MuSiQue	R5−R1	1744	-0.011	[-0.026, +0.003]	0.122	0.535	68/88	522	-0.016	+0.003	no
MuSiQue	R4−R5	1744	+0.010	[-0.003, +0.024]	0.117	0.535	75/57	0	+0.010	+0.004	no
2Wiki	R1−R0	3232	-0.017	[-0.029, -0.005]	0.003	0.018	185/239	44	-0.017	-0.052	no
2Wiki	R5−R1	3232	-0.008	[-0.018, +0.002]	0.107	0.535	126/152	1858	-0.019	-0.012	yes
2Wiki	R4−R5	3232	+0.006	[-0.002, +0.015]	0.174	0.535	103/84	879	+0.008	-0.011	yes

Table 2: Confirmatory test of the six pre-specified primary contrasts (Protocol A, EM, disjoint sample; `docs/CONFIRMATORY_PROTOCOL.md`). p: two-sided paired bootstrap; p_{Holm}: Holm-adjusted over the six tests. Equiv.: 90% CI inside ± 0.02 (pre-specified secondary check). Source: `F_exp1_contrasts.csv`.

($+0.002$, $[-0.013, +0.017]$; Appendix Table 12). Derived relation lines show the mirror image: they raise accuracy among answered questions ($+0.014$, $[+0.005, +0.024]$) but also abstention, and the two cancel.

6.2 Fixed-order controls: the asserted label carries the cost (exploratory)

The secondary contrast R4P−R4, predicted relations instead of derived ones, costs 3.6 (2WikiMultiHopQA) and 4.5 (MuSiQue) points, but changes order, line count and labels together. Table 3 changes them one at a time on the same questions. Showing the predicted `CONTRADICTS` lines rather than removing them costs 3.3 points on both datasets and raises abstention by 6 points. Two tighter interventions isolate the word itself. Relabelling only those lines as `RELATED_TO`, with every pair, position and other label unchanged, recovers as much (3.7 and 3.3 points); so does replacing each with a filler line of the same count and nearly the same length (4.2 and 3.7 points). Relabelling the other predicted labels adds nothing beyond relabelling the contradictions (0.000 and $+0.003$), the predicted order alone changes at most 1.8 points, and in blocks without contradiction lines the prompts are identical and the effect is zero by construction. Among questions with at least one such line, the lines cost 5–7 points. Relabelling restored 141 correct 2WikiMultiHopQA answers (80 from abstentions) and lost 21 (MuSiQue: 62 and 4).

6.3 Confirmatory test 2: contradiction lines on untouched questions

Table 4 reports the pre-specified test of R4P_FO against R4P_FO_NC on confirmatory sample 2. Both hypotheses are confirmed. Among questions whose block shows a contradiction line (1,490 of 2,600 in 2WikiMultiHopQA, 1,295 of 3,800 in HotpotQA), showing the lines lowers exact match by 7.7 points $[-9.5, -6.1]$ and 5.3 points $[-6.7, -3.9]$ ($p_{\text{Holm}} < 0.001$ for both) and raises abstention by 8.9 and 7.6 points. Over all questions the cost is 4.4 and 1.8 points. Abstention is the main channel but not the only one: about half of the lost correct answers become abstentions and half become wrong answers, and accuracy among questions answered under both conditions also falls (by 5.5 and 2.3 points).

6.4 A second reader (exploratory)

Appendix Table 14 repeats the main contrasts with Mistral-7B-Instruct-v0.3 on the confirmatory questions. The contradiction effect replicates: showing the lines at a fixed order costs Mistral 3.4 (2WikiMultiHopQA) and 1.5 (MuSiQue) points of exact match, 5.0 and 3.3 points of lenient match, with 5–6 points more abstention, and predicted lines cost more than derived ones on both datasets. The sentence-item effect does not replicate: for Mistral, R1−R0 is -0.001 on 2WikiMultiHopQA and $+0.022$ $[+0.012, +0.033]$ on MuSiQue, where sentence items reduce its abstention by 4.6 points. Derived order and relation lines remain near zero for both readers. The cost of asserted contradictions is therefore shared by two model families, whereas the effect of segmentation depends on the reader.

6.5 Retrieved evidence and abstention policy (exploratory)

On 1,000 fresh questions where every representation renders the same four reranked passages (Appendix Tables 20–21), predicted relations cost 3.6 (MuSiQue) and 3.0 (2WikiMultiHopQA) points and contradiction lines 2.2 and 2.4; the costs concentrate where retrieval returned all supporting sentences and vanish when it returned none. Annotation-derived edges rarely survive retrieval, so correct structure on incomplete evidence remains untested. Varying the abstention policy

Factor changed (first − second)	2Wiki ΔEM	95% CI	ΔAbst.	Ident.	MuSiQue ΔEM	95% CI	ΔAbst.	Ident.
contradiction lines (on vs. off), derived order	-0.033	[-0.042, -0.025]	+0.061	1198	-0.033	[-0.043, -0.023]	+0.062	734
contradiction label vs. neutral label, those lines only	-0.037	[-0.045, -0.029]	+0.058	1198	-0.033	[-0.042, -0.025]	+0.054	734
contradiction lines vs. equal-count filler lines	-0.042	[-0.050, -0.034]	+0.067	1198	-0.037	[-0.047, -0.028]	+0.070	734
typed vs. neutral labels for all lines, same pairs	-0.037	[-0.045, -0.029]	+0.056	530	-0.030	[-0.040, -0.020]	+0.048	410
neutral lines vs. none (count/overhead only)	+0.007	[-0.003, +0.017]	+0.019	530	-0.015	[-0.026, -0.004]	+0.038	410
predicted vs. derived order, no lines	+0.002	[-0.001, +0.005]	-0.003	3057	+0.018	[+0.006, +0.031]	+0.003	704
predicted vs. derived order, predicted lines	+0.000	[-0.003, +0.003]	-0.003	3057	+0.010	[-0.002, +0.023]	-0.001	704
original R4P−R4 (order and lines)	-0.036	[-0.045, -0.027]	+0.061	818	-0.045	[-0.060, -0.029]	+0.089	94

Table 3: Fixed-order relation controls on the confirmatory questions (exploratory interventions designed after the confirmatory result; n=3,232 2Wiki, 1,744 MuSiQue). Each row changes one factor on identical facts. ΔAbst.: paired change in the abstention rate; Ident.: byte-identical prompt pairs (questions without contradiction lines for rows 1–3). Sources: `revision/controls/relation_control_effects.csv`, `revision2/r2/controls2.csv`.

Data	n (lines/all)	ΔEM	95% CI	p_{Holm}	ΔAbst.	95% CI	ΔEM all	ΔAcc|both	C→A/C→W/A→C/W→C
2Wiki	1,490/2,600	-0.077	[-0.095, -0.061]	<0.001	+0.089	[+0.072, +0.104]	-0.044	-0.055	67/73/4/21
HotpotQA	1,295/3,800	-0.053	[-0.067, -0.039]	<0.001	+0.076	[+0.061, +0.092]	-0.018	-0.023	46/37/1/13

Table 4: Confirmatory test 2 (protocol frozen and published before sampling): showing the predicted `CONTRADICTS` lines (R4P_FO) vs. the same block without them (R4P_FO_NC), fixed order, on untouched questions. Primary population: questions with at least one rendered contradiction line (an input property); Holm over the two datasets. ΔEM all: secondary estimand over all questions (identical prompts contribute 0). ΔAcc|both and transitions are descriptive. Source: `revision2/confirm2/confirm2_summary.json`.

(Appendix Table 15) changes every representation's level but not the contradiction cost: under stricter guidance predicted relations still cost 3.3 points; without the abstention option the differences shrink (-0.004 and -0.017), but wrong answers rise from 33–40% to 51–52%.

7 Discussion

Reorganising the same facts changed little. With the facts held fixed, neither the graph-derived order nor explicit derived relation lines changed exact match by more than about one point, for either reader. Gains attributed to graph-structured evidence in system-level comparisons are thus unlikely to come from relation lines or ordering as such.

When structure hurts: asserted contradictions. The clearest and only confirmed harm came from relation lines that assert a conflict. Gold supporting passages are, by construction, consistent evidence for one answer, yet the NLI classifier marks at least one pair as contradictory in 34–63% of prompts. The interventions show that the word itself does the damage: relabelling these lines, or blanking them while keeping their number and length, recovers the loss, and nothing else about the predicted block matters measurably. The effect held prospectively on untouched questions and a new dataset, and for a second model family; the cost of sentence items, by contrast, was reader-specific (for Mistral it was null or positive). Complementing Nishida et al. (2026), who show that noisy relations hurt GraphRAG through retrieval, they also hurt when merely displayed next to fixed, correct evidence. Automatic triage suggests most such pairs are false (40% concern different entities, 20% different dates), but we establish only the cost of *showing* an asserted contradiction; whether it requires falsity awaits human labels.

Abstention is the main channel, not the only one. Most effects ran through the reader's willingness to answer, but asserted contradictions also produced wrong answers on fresh questions, and removing the abstention option is no remedy (12–19 more points of wrong answers). Accuracy alone conflates caution with reasoning failures; we recommend reporting outcome transitions for any comparison of evidence formats.

8 Conclusion

With the evidence fixed, we found no detectable effect of graph-derived order or derived relation lines on 4,976 held-out questions, and a small, reader-specific cost of sentence items. What hurt was an asserted contradiction: with facts and order fixed, showing NLI-predicted `CONTRADICTS` lines lowered exact match by 5–8 points among the affected questions in a pre-specified test on untouched questions, mainly through abstention, and the cost vanished when only the label was changed. Evidence structure should be judged by what its annotations make a reader do.

Limitations

Two 7B readers, one prompt family. All results come from Qwen2.5-7B-Instruct, with Mistral-7B-Instruct-v0.3 as an exploratory second reader on the same questions, both with greedy decoding. Two readers cannot establish behaviour of model families in general, and larger or differently trained models may treat asserted contradictions differently. Most effects act through abstention under a prompt that offers *insufficient evidence*; without that option the differences shrink but wrong answers rise.

Asserted, not verified, falsity. The confirmed result concerns showing NLI-predicted contradiction lines. That most of them are false rests on automatic triage; the blinded two-annotator package is prepared but not labelled, so we report no human-verified precision, and we did not run the intervention that removes only verified false contradictions. Every derived dependency edge is derived by rule, and a substantial fraction has a fallback anchor; no relation is human-verified, and the component-attribution experiment was not run.

Sentence units, not propositions. Our fact units are dataset sentences, some of which depend on their passage, not LLM-extracted atomic propositions. The results concern sentence-item presentation, not the end-to-end effect of claim extraction, which we observed only at pilot scale with unequal budgets.

Gold evidence and retrieval. The confirmatory tests and controls use gold supporting passages. The retrieved-evidence experiment uses 500 questions per dataset, its MuSiQue sample is almost entirely 2-hop, and annotation-derived structure rarely survives top-4 retrieval, so correct structure on incomplete evidence remains untested.

Exploratory analyses. The fixed-order controls, the prompt variants, the second reader and the retrieved-evidence experiment were designed after the first confirmatory result and are not multiplicity-corrected; only the contradiction contrast was tested prospectively. A planned regression of the contradiction effect on line counts was abandoned because its predictors are collinear.

Metric and pilot scale. Exact match is strict, especially for a reader that answers in sentences; we therefore also report a lenient match fixed before that reader's run. Protocol B, Experiment 0 and the perturbations were run only at pilot scale, and the pilot overestimated the first confirmed effect by a factor of four.

Compute. All runs used one 32 GB GPU with batch size 1, about 20 GPU-hours in total.

Ethical Considerations

All questions and passages come from public research datasets (MuSiQue, 2WikiMultiHopQA, HotpotQA, HoVer, FEVER) and are used under their licences for research; we release no new text data. No human subjects were involved in the reported results. The blinded annotation packages we prepared will, if used, require informed, paid annotators and are described in the repository. All models are openly licensed and were run locally on one GPU; the full study used about 20 GPU-hours. Our results concern how a reader responds to evidence formats, including a tendency to abstain when shown asserted conflicts; they do not make systems more capable of harm, but they do suggest that automatically extracted relations can silently reduce the answers a deployed system gives. We used an AI assistant extensively for code, experiment execution and drafting; the scope of this use is disclosed in the Responsible NLP checklist, and the authors are responsible for all content.

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A Pipeline and prompts

Every Experiment 0–2 answer uses the same prompt (version `qa-v2`); only the evidence block differs between representations. The system message is *You answer using only the evidence you are given.* The user message is:

```
Use only the evidence below to answer the question.
Each evidence item has an ID such as [E1]. Items may
be grouped into steps or followed by a list of
relations between items.

Evidence:
{evidence}

Question: {question}

Reply in exactly two lines:
Answer: <short answer: a name, date, number, short
phrase, or yes/no>
Evidence: <IDs of the items you used, e.g. E1, E2>
If the evidence is not sufficient, reply
"Answer: insufficient evidence".
```

The assistant turn is prefilled with `Answer:`. HoVer and FEVER use the same template with *Claim* in place of *Question*, ask for one of the dataset's labels, and have no abstention line. The claim extractor of Experiment 0 (version `extract-v2`) uses the system message *You split sentences into short, simple facts.* and asks for one fact per line, naming the subject instead of using pronouns and using only information stated in the sentence. Its output is prefilled with `-` .

B Serialisation examples

The following renderings are produced by the experiment code for a MuSiQue development question that was used only during prompt development and is excluded from every reported run: *What character in Princess Diaries does the actress play who was also a cast member in On Golden Pond?* (gold: *Queen dowager Clarisse Renaldi*). Long sentences are abbreviated here with "…".

R0 (passages).

```
[E1] (source: The Princess Diaries
(film)) The Princess Diaries is a 2001
American teen romantic comedy ... It
stars Anne Hathaway ... ruled by her
grandmother Queen dowager Clarisse
Renaldi (Julie Andrews). ...
[E2] (source: On Golden Pond (2001
film)) On Golden Pond is a 2001
American television adaptation of the
play starring Julie Andrews and
Christopher Plummer. ...
```

R1 (sentences, dataset order).

```
[E1] (source: The Princess Diaries
(film)) The Princess Diaries is a 2001
American teen romantic comedy ...
[E2] (source: The Princess Diaries
(film)) It stars Anne Hathaway ...
Queen dowager Clarisse Renaldi (Julie
Andrews).
[E3] (source: The Princess Diaries
(film)) The film also stars ...
[E4] (source: On Golden Pond (2001
film)) On Golden Pond is a 2001
American television adaptation of the
play starring Julie Andrews and
Christopher Plummer.
[E5] ... [E6] ...
```

Item E2 begins with a pronoun. Separated from its passage, its subject is recoverable only through the source title that every item carries. We record such context-dependent facts per question; they are one

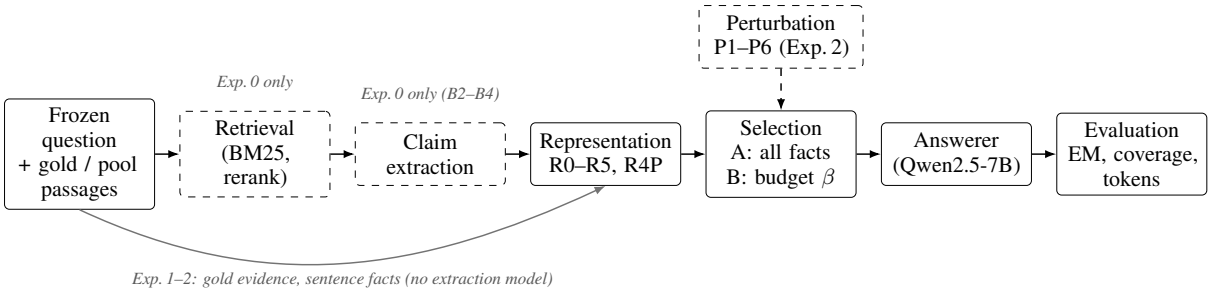


Figure 1: The evidence pipeline. Experiment 1 starts from the gold passages and varies only the representation (Protocol A), or the representation under a fixed fact ranking and token budget (Protocol B). Experiment 0 runs the retrieved-evidence pipeline for five systems; Experiment 2 corrupts the evidence of two representations. Dashed boxes are used only by some conditions.

way in which an information-equivalent transformation can still change what is easy to read.

R3 (steps), R5 (graph order) and R4 (graph order plus relations). R3 groups the same six items under `Step 1:` (the three *On Golden Pond* sentences) and `Step 2:` (the three *Princess Diaries* sentences), following the MuSiQue decomposition. R5 lists the same items in the same order without step headers. R4 is R5 followed by

```
Relations:
E5 DEPENDS_ON E1
```

i.e. the sentence naming the character depends on the sentence establishing that Julie Andrews was in *On Golden Pond*. This edge is derived from the MuSiQue decomposition with an exact anchor; it is not human-verified.

C Reconstruction of the EG-RAG baseline

B4 follows the stage order of the EG-RAG algorithm: hybrid retrieval, reranking, claim extraction, relation prediction, belief and utility propagation, conflict resolution, budgeted subgraph selection and serialisation. Every component that the original description leaves unspecified is our choice:

- BM25 plus bge-small-en-v1.5 with min–max normalisation and equal weights;
- ms-marco-MiniLM-L6-v2 reranking;
- the 7B generator as claim extractor, with a lexical-support filter;
- nli-deberta-v3-xsmall for support, contradiction and duplication, with thresholds 0.7 and 0.9;
- a title-mention heuristic for dependencies;
- years as the only temporal signal;
- our own feature weights, a 450-token budget and at most 12 claims;
- greedy connected selection.

The full ledger is in the repository (`docs/NOVELTY_AND_BASELINES.md`). B4 is therefore a reconstruction under our choices. Its results say nothing about the original implementation, and no result of the original paper is reused.

D Perturbations

Experiment 2 applies each intervention to the R1 and R4 renderings of the gold evidence with a recorded seed (0). The rates are $\{0.2, 0.4\}$ for P1–P4 and 0.4 for P5 and P6, each against the same question's rate-0 record, and $k = \lceil \text{rate} \cdot n_{\text{eligible}} \rceil$ items are affected.

- **P1** removes required (gold-support) facts.
- **P2** retypes or flips derived relations (R4 only).
- **P3** permutes item positions.
- **P4** inserts real non-gold sentences from the question's own candidate pool, chosen by BM25 similarity (*hard*), and never sentences containing a gold answer.
- **P5** strips source titles.
- **P6** deletes a negation, or replaces an entity in a fact with *it*.

Every changed item is recorded before and after. P1 and P6 can remove the answer and are

therefore not label-preserving; we report their answer-change and abstention rates but no accuracy change. An intervention is unavailable when nothing is eligible:

- P2 on R1 or on questions without edges;
- P3 with a single item;
- P4 on HoVer and FEVER, which have no candidate pool;
- P6 when no fact contains a negation;
- P7 everywhere, since no verified temporal data exist.

These cases are counted and never scored as zero.

E Details of the exploratory pilot results

Evidence loss under a token budget (pilot). Protocol B was run only in the exploratory pilot (Figure 3 and Table 9, Appendix F). Its accounting of what each condition showed is deterministic and does not depend on the generator. Under a budget equal to the content tokens of the fact set, with each representation keeping as much of the shared ranking as it can afford, R0 keeps 0.83 (2WikiMultiHopQA) and 0.81 (MuSiQue) of the facts, whereas R1 keeps 0.64 and 0.65 and R4 keeps 0.63 and 0.62, because their identifiers, titles and relation lines cost 104–108 (2Wiki) and 81–88 (MuSiQue) tokens of overhead against 28 and 32 for R0. The verbose formats also lose the facts that matter: R0 drops 0.11 answer-bearing facts per question on 2WikiMultiHopQA against 0.33 for R1 and 0.34 for R4 (MuSiQue: 0.19, 0.28 and 0.35), so that the answer string remains in 0.85 of R0 blocks but only 0.66 of R1 and 0.64 of R4 blocks. The corresponding accuracy differences are smaller than these evidence losses and inconclusive at $n = 100$. For example, R1−R0 is -0.05 $[-0.11, +0.01]$ on 2WikiMultiHopQA, and R4−R5 is -0.05 $[-0.11, +0.01]$ on MuSiQue. Under the shared policy, every representation shows the same facts and pays the same evidence loss. Halving the budget reduces exact match for every representation to 0.08–0.19.

Retrieved-evidence baselines (pilot). Table 10 (Appendix F) reports Experiment 0 on the 200 primary pilot questions. Reranking (B1) improves on BM25 (B0) on MuSiQue by $+0.06$ $[0.00, +0.12]$, and the 2WikiMultiHopQA difference is inconclusive. Extracted claims (B2) and predicted reasoning paths (B3) are indistinguishable from reranked passages in exact match; the extracted claims keep the answer string as often as the passages they come from (0.78 and 0.44), but at roughly three times the evidence tokens (1,390–1,711 against 501–538), because every passage sentence becomes several facts. Our reconstruction of EG-RAG (B4) uses a similar number of tokens to B1 (442–489 against 501–538) but is less accurate, by -0.15 $[-0.23, -0.07]$ on 2WikiMultiHopQA and -0.09 $[-0.15, -0.04]$ on MuSiQue: its budgeted selection keeps the answer string less often (0.56 against 0.78; 0.31 against 0.44), and coverage falls by 0.29 and 0.15.

Controlled perturbations (pilot). Table 11 and Figure 2 (Appendix F) summarise Experiment 2. These comparisons are many (roughly fifty cells), and we expect a few intervals to exclude zero by chance; we report the patterns that recur across datasets or representations. When required facts are removed (P1), the model mostly abstains rather than guessing: abstention rises from 0.43 to 0.72 on MuSiQue (R1) and from 0.24 to 0.48 on 2WikiMultiHopQA. Hard distractors (P4) reduce exact match in every QA cell, by up to 0.16 (HotpotQA R1, $[-0.28, -0.04]$) and by 0.09 on MuSiQue R1 ($[-0.17, -0.01]$), and raise abstention on MuSiQue. Retyping or flipping 40% of the derived relations (P2, R4 only) lowers 2WikiMultiHopQA exact match by 0.10 $[-0.18, -0.01]$ on the 71 questions that have relations, and is inconclusive on MuSiQue and HotpotQA. Reordering (P3) and stripping source titles (P5) produce small effects of inconsistent sign, and HoVer and FEVER verdicts are almost unaffected by any intervention.

On the secondary datasets (50 questions each; run `C_exp0`), every Experiment 0 contrast is inconclusive, e.g. B4−B1 is 0.00 $[-0.12, +0.12]$ EM on HotpotQA and -0.06 $[-0.22, +0.12]$ verdict accuracy on HoVer.

F Additional results

Table 8 gives the absolute scores of every representation under Protocol A in the expanded pilot, Table 9 the full information accounting under Protocols A and B, Table 10 the retrieved-evidence baselines, and Table 11 and Figure 2 the perturbation experiment. Table 5 lists the secondary contrasts

on the confirmatory sample and Table 7 the analysis of predicted contradiction lines; the remaining tables and figures are exploratory pilot results.

G Revision experiments: supplementary tables

Table 12 gives the outcome transitions behind the first confirmatory test and the controls, and Figure 4 the revision-1 controls graphically. Table 13 stratifies the contradiction-line effect by the number of such lines. Table 14 gives the second-reader contrasts, and Table 15 compares the three abstention policies. Tables 16, 20 and 21 report the retrieved-evidence experiment. Table 17 accounts for the tokens, facts, relation lines and source titles of every representation, and Table 18 splits the derived-structure contrasts by anchor quality. Table 19 summarises the predicted contradictions, and Table 22 makes the budgets and costs of Experiment 0 explicit. All of these are exploratory. The blinded two-annotator verification package for 300 predicted contradictions and its importer are described in the repository.

Data	Contrast	n	ΔEM	95% CI	Ident.	ΔCov.
MuSiQue	R2−R1	1744	+0.013	[-0.001, +0.028]	2	+0.041
MuSiQue	R3−R5	1744	-0.007	[-0.016, +0.002]	0	-0.015
MuSiQue	R4P−R4	1744	-0.045	[-0.060, -0.030]	94	-0.089
2Wiki	R2−R1	3232	-0.024	[-0.036, -0.012]	4	-0.017
2Wiki	R3−R5	3232	-0.004	[-0.016, +0.008]	0	-0.015
2Wiki	R4P−R4	3232	-0.036	[-0.045, -0.027]	818	-0.061

Table 5: Secondary contrasts on the confirmatory sample (pre-specified as secondary; not part of the tested family, so descriptive). Source: `F_exp1_contrasts.csv`.

Data	Contrast	ΔEM	both ans.	one abstains	abst. only first / only second	neither
2Wiki	R1−R0	-0.017	+0.001	-0.018	250/81	529
2Wiki	R4−R5	+0.006	+0.010	-0.004	123/88	731
2Wiki	R4P−R4	-0.036	-0.011	-0.025	257/59	795
2Wiki	R5−R1	-0.008	-0.004	-0.004	170/130	649
MuSiQue	R1−R0	-0.001	-0.002	+0.001	94/110	710
MuSiQue	R4−R5	+0.010	+0.006	+0.004	82/89	709
MuSiQue	R4P−R4	-0.045	-0.005	-0.040	218/63	728
MuSiQue	R5−R1	-0.011	-0.012	+0.001	98/104	700

Table 6: Exploratory decomposition of each paired EM difference on the confirmatory sample (added after the confirmatory result). *both ans.*: contribution of questions answered under both representations; *one abstains*: contribution of questions answered under only one. The columns sum to ΔEM. Source: `F_exp1_decomposition_exploratory.csv`.

Data	R4P shows CONTRADICTS	n	Abst. R4	Abst. R4P	EM R4	EM R4P
2Wiki	yes	2034	0.32	0.41	0.33	0.28
2Wiki	no	1198	0.17	0.17	0.48	0.48
MuSiQue	yes	1010	0.50	0.63	0.25	0.18
MuSiQue	no	734	0.39	0.42	0.33	0.32

Table 7: Exploratory: the same questions under derived relations (R4) and predicted relations (R4P), split by whether the predicted block contains at least one `CONTRADICTS` line (an input property). Source: `F_exp1_r4p_relations_exploratory_paired.csv`.

Data	R0	R1	R2	R3	R4	R4P	R5
MuSiQue	0.28 (0.57)	0.31 (0.57)	0.28 (0.59)	0.24 (0.52)	0.29 (0.59)	0.21 (0.42)	0.29 (0.58)
2Wiki	0.44 (0.84)	0.37 (0.76)	0.36 (0.74)	0.37 (0.67)	0.41 (0.69)	0.38 (0.66)	0.38 (0.70)
HotpotQA	0.40 (0.88)	0.42 (0.82)	0.35 (0.85)	0.40 (0.86)	0.40 (0.84)	0.42 (0.80)	0.38 (0.82)
HoVer	0.68 (1.00)	0.68 (1.00)	0.73 (1.00)	–	0.68 (1.00)	0.74 (1.00)	0.68 (1.00)
FEVER	0.96 (1.00)	0.96 (1.00)	0.97 (1.00)	–	0.96 (1.00)	0.96 (1.00)	0.96 (1.00)

Table 8: Exploratory pilot, Protocol A: EM (QA) or verdict accuracy (HoVer, FEVER), with coverage (answered rate) in parentheses, for the same facts in seven representations. n=100 (MuSiQue, 2Wiki) or 50. R3 is unavailable without a reasoning order. Source: `C_exp1_absolute.csv`.

Data	Setting	Rep.	Kept	Ovh.	Tok.	Ans. drop	Ans. in ev.	EM	Cov.
MuSiQue	A	R0	1.00	33	378	0.00	0.99	0.28	0.57
MuSiQue	A	R1	1.00	122	467	0.00	0.99	0.31	0.57
MuSiQue	A	R5	1.00	122	467	0.00	0.99	0.29	0.58
MuSiQue	A	R4	1.00	141	486	0.00	0.99	0.29	0.59
MuSiQue	B 1.0× per-rep.	R0	0.81	32	327	0.19	0.89	0.25	0.52
MuSiQue	B 1.0× per-rep.	R1	0.65	81	322	0.28	0.85	0.26	0.52
MuSiQue	B 1.0× per-rep.	R5	0.65	81	322	0.28	0.85	0.22	0.53
MuSiQue	B 1.0× per-rep.	R4	0.62	88	321	0.35	0.80	0.17	0.44
MuSiQue	B 1.0× shared	R0	0.60	30	251	0.38	0.77	0.18	0.47
MuSiQue	B 1.0× shared	R1	0.60	75	296	0.38	0.77	0.20	0.48
MuSiQue	B 1.0× shared	R5	0.60	75	296	0.38	0.77	0.18	0.52
MuSiQue	B 1.0× shared	R4	0.60	86	307	0.38	0.77	0.17	0.46
MuSiQue	B 0.5× per-rep.	R0	0.34	23	144	0.71	0.49	0.10	0.30
MuSiQue	B 0.5× per-rep.	R1	0.30	38	143	0.77	0.43	0.09	0.30
MuSiQue	B 0.5× per-rep.	R5	0.30	38	143	0.77	0.43	0.09	0.30
MuSiQue	B 0.5× per-rep.	R4	0.29	39	142	0.79	0.42	0.08	0.31
MuSiQue	B 0.5× shared	R0	0.27	21	120	0.80	0.42	0.09	0.28
MuSiQue	B 0.5× shared	R1	0.27	35	133	0.80	0.42	0.09	0.30
MuSiQue	B 0.5× shared	R5	0.27	35	133	0.80	0.42	0.09	0.30
MuSiQue	B 0.5× shared	R4	0.27	38	136	0.80	0.42	0.08	0.30
2Wiki	A	R0	1.00	29	414	0.00	0.98	0.44	0.84
2Wiki	A	R1	1.00	157	543	0.00	0.98	0.37	0.76
2Wiki	A	R5	1.00	157	543	0.00	0.98	0.38	0.70
2Wiki	A	R4	1.00	167	552	0.00	0.98	0.41	0.69
2Wiki	B 1.0× per-rep.	R0	0.83	28	366	0.11	0.85	0.38	0.78
2Wiki	B 1.0× per-rep.	R1	0.64	104	361	0.33	0.66	0.33	0.67
2Wiki	B 1.0× per-rep.	R5	0.64	104	361	0.33	0.66	0.31	0.64
2Wiki	B 1.0× per-rep.	R4	0.63	108	361	0.34	0.64	0.33	0.64
2Wiki	B 1.0× shared	R0	0.60	26	265	0.39	0.60	0.30	0.67
2Wiki	B 1.0× shared	R1	0.60	97	337	0.39	0.60	0.27	0.63
2Wiki	B 1.0× shared	R5	0.60	97	337	0.39	0.60	0.27	0.62
2Wiki	B 1.0× shared	R4	0.60	103	342	0.39	0.60	0.29	0.62
2Wiki	B 0.5× per-rep.	R0	0.39	21	173	0.59	0.43	0.17	0.50
2Wiki	B 0.5× per-rep.	R1	0.32	52	171	0.62	0.39	0.17	0.39
2Wiki	B 0.5× per-rep.	R5	0.32	52	171	0.62	0.39	0.19	0.41
2Wiki	B 0.5× per-rep.	R4	0.31	52	170	0.62	0.39	0.15	0.38
2Wiki	B 0.5× shared	R0	0.29	20	131	0.63	0.38	0.14	0.45
2Wiki	B 0.5× shared	R1	0.29	48	160	0.63	0.38	0.15	0.36
2Wiki	B 0.5× shared	R5	0.29	48	160	0.63	0.38	0.16	0.37
2Wiki	B 0.5× shared	R4	0.29	50	161	0.63	0.38	0.15	0.37

Table 9: Information accounting under Protocol A and Protocol B (budget $\beta\times$ the content tokens of the fact set). Kept: fraction of facts shown; Ovh./Tok.: mean overhead and evidence tokens; Ans. drop: answer-bearing facts cut per question; Ans. in ev.: answer string present in the block. Source: `C_exp1_protocol_accounting.csv`, `C_exp1_absolute.csv`.

Data	Sys.	EM	95% CI	F1	Cov.	Tok.	Units	Ans. in ev.	Lat. (s)
MuSiQue	B0	0.06	[+0.02, +0.11]	0.10	0.23	493	4.0	0.34	0.36
MuSiQue	B1	0.12	[+0.06, +0.19]	0.17	0.33	538	4.0	0.44	0.36
MuSiQue	B2	0.13	[+0.07, +0.20]	0.17	0.36	1390	57.1	0.44	0.86
MuSiQue	B3	0.15	[+0.08, +0.22]	0.18	0.38	1667	57.1	0.44	0.91
MuSiQue	B4	0.03	[+0.00, +0.07]	0.05	0.18	442	11.9	0.31	0.51
2Wiki	B0	0.23	[+0.16, +0.31]	0.27	0.54	506	4.0	0.69	0.38
2Wiki	B1	0.26	[+0.18, +0.35]	0.30	0.54	501	4.0	0.78	0.39
2Wiki	B2	0.24	[+0.16, +0.33]	0.27	0.48	1436	56.4	0.78	0.67
2Wiki	B3	0.20	[+0.12, +0.28]	0.24	0.50	1711	56.4	0.78	0.68
2Wiki	B4	0.11	[+0.05, +0.17]	0.13	0.25	489	11.9	0.56	0.51

Table 10: Experiment 0, retrieved evidence (exploratory, n=100 per dataset). Tok.: mean evidence tokens; Units: items shown; Lat.: answer-generation latency (claim extraction for B2–B4 is not included). B4 is our reconstruction of EG-RAG. Source: `A_exp0_absolute.csv`.

Data	Rep.	Perturbation	n	Lab.-pres.	ΔEM	95% CI	Ans. changed	Abstention
MuSiQue	R1	P1	100	0	–	–	0.40	0.43→0.72
MuSiQue	R1	P3	100	100	-0.07	[-0.13, -0.02]	0.13	0.43→0.44
MuSiQue	R1	P4 hard	100	100	-0.09	[-0.17, -0.01]	0.27	0.43→0.51
MuSiQue	R1	P5 strip	100	100	-0.03	[-0.08, +0.02]	0.12	0.43→0.41
MuSiQue	R1	P6 entity	100	0	–	–	0.14	0.43→0.50
MuSiQue	R1	P6 negation	29	0	–	–	0.03	0.52→0.55
MuSiQue	R4	P1	100	0	–	–	0.44	0.41→0.71
MuSiQue	R4	P2	100	100	-0.01	[-0.07, +0.05]	0.24	0.41→0.47
MuSiQue	R4	P3	100	100	-0.03	[-0.08, +0.02]	0.14	0.41→0.44
MuSiQue	R4	P4 hard	100	100	-0.08	[-0.16, +0.00]	0.34	0.41→0.54
MuSiQue	R4	P5 strip	100	100	+0.00	[-0.04, +0.04]	0.10	0.41→0.38
MuSiQue	R4	P6 entity	100	0	–	–	0.19	0.41→0.43
MuSiQue	R4	P6 negation	29	0	–	–	0.03	0.48→0.52
2Wiki	R1	P1	100	0	–	–	0.43	0.24→0.48
2Wiki	R1	P3	100	100	+0.04	[-0.04, +0.12]	0.30	0.24→0.19
2Wiki	R1	P4 hard	100	100	-0.04	[-0.13, +0.05]	0.35	0.24→0.31
2Wiki	R1	P5 strip	100	100	+0.02	[-0.03, +0.08]	0.16	0.24→0.18
2Wiki	R1	P6 entity	100	0	–	–	0.14	0.24→0.27
2Wiki	R1	P6 negation	15	0	–	–	0.00	0.20→0.20
2Wiki	R4	P1	100	0	–	–	0.43	0.31→0.51
2Wiki	R4	P2	71	71	-0.10	[-0.18, -0.01]	0.44	0.35→0.34
2Wiki	R4	P3	100	100	-0.01	[-0.09, +0.07]	0.26	0.31→0.26
2Wiki	R4	P4 hard	100	100	-0.03	[-0.12, +0.06]	0.35	0.31→0.36
2Wiki	R4	P5 strip	100	100	+0.08	[+0.01, +0.16]	0.19	0.31→0.26
2Wiki	R4	P6 entity	100	0	–	–	0.17	0.31→0.33
2Wiki	R4	P6 negation	15	0	–	–	0.00	0.40→0.40
HotpotQA	R1	P1	50	0	–	–	0.36	0.18→0.44
HotpotQA	R1	P3	50	50	-0.06	[-0.16, +0.04]	0.18	0.18→0.14
HotpotQA	R1	P4 hard	50	50	-0.16	[-0.28, -0.04]	0.26	0.18→0.24
HotpotQA	R1	P5 strip	50	50	-0.02	[-0.08, +0.04]	0.08	0.18→0.14
HotpotQA	R1	P6 entity	50	0	–	–	0.18	0.18→0.18
HotpotQA	R1	P6 negation	5	0	–	–	0.00	0.20→0.20
HotpotQA	R4	P1	50	0	–	–	0.42	0.16→0.44
HotpotQA	R4	P2	25	25	-0.04	[-0.24, +0.12]	0.28	0.08→0.16
HotpotQA	R4	P3	50	50	-0.04	[-0.14, +0.06]	0.14	0.16→0.16
HotpotQA	R4	P4 hard	50	50	-0.10	[-0.24, +0.04]	0.34	0.16→0.20
HotpotQA	R4	P5 strip	50	50	+0.02	[-0.04, +0.08]	0.08	0.16→0.14
HotpotQA	R4	P6 entity	50	0	–	–	0.16	0.16→0.18
HotpotQA	R4	P6 negation	5	0	–	–	0.00	0.20→0.20

Table 11: Experiment 2 at rate 0.4 (exploratory). ΔEM is computed only on label-preserving records without introduced answers (Lab.-pres.); P1 and P6 can remove the answer, so only answer-change and abstention rates are given. Source: `C_exp2_sensitivity.csv`.

Data	Change (second vs. first)	n	ΔEM	ΔAbst.	ΔWrong	ΔAcc. | both answered	C→A / A→C	C→W / W→C	W→A / A→W
2Wiki	sentence items vs. passages	3232	-0.017	+0.052	-0.036	+0.002 [-0.013, +0.017]	87/29	152/156	163/52
2Wiki	graph order vs. dataset order	3232	-0.008	+0.012	-0.004	-0.006 [-0.018, +0.005]	57/45	95/81	113/85
2Wiki	derived relation lines vs. none	3232	+0.006	+0.011	-0.017	+0.014 [+0.005, +0.024]	42/29	42/74	81/59
2Wiki	R4P vs. R4 (original)	3232	-0.036	+0.061	-0.025	-0.017 [-0.027, -0.007]	97/17	80/44	160/42
2Wiki	contradiction lines on vs. off	3232	-0.033	+0.061	-0.028	-0.013 [-0.022, -0.004]	89/9	64/36	137/20
MuSiQue	sentence items vs. passages	1744	-0.001	-0.009	+0.010	-0.004 [-0.020, +0.014]	42/44	29/26	52/66
MuSiQue	graph order vs. dataset order	1744	-0.011	-0.003	+0.015	-0.025 [-0.045, -0.006]	45/46	43/22	53/58
MuSiQue	derived relation lines vs. none	1744	+0.010	-0.004	-0.006	+0.013 [-0.007, +0.031]	28/35	29/40	54/54
MuSiQue	R4P vs. R4 (original)	1744	-0.045	+0.089	-0.044	-0.012 [-0.037, +0.011]	93/24	43/34	125/39
MuSiQue	contradiction lines on vs. off	1744	-0.033	+0.062	-0.029	-0.009 [-0.020, +0.003]	52/2	14/7	64/6

Table 12: Question-level outcome transitions (C correct, W answered and wrong, A not answered) between paired conditions on the confirmatory questions (exploratory decomposition). ΔAcc. | both answered: accuracy difference among questions answered under both conditions, with 95% CI. Sources: `revision/analysis/F_exp1_transitions.csv`, `revision/controls/controls_transitions.csv`.

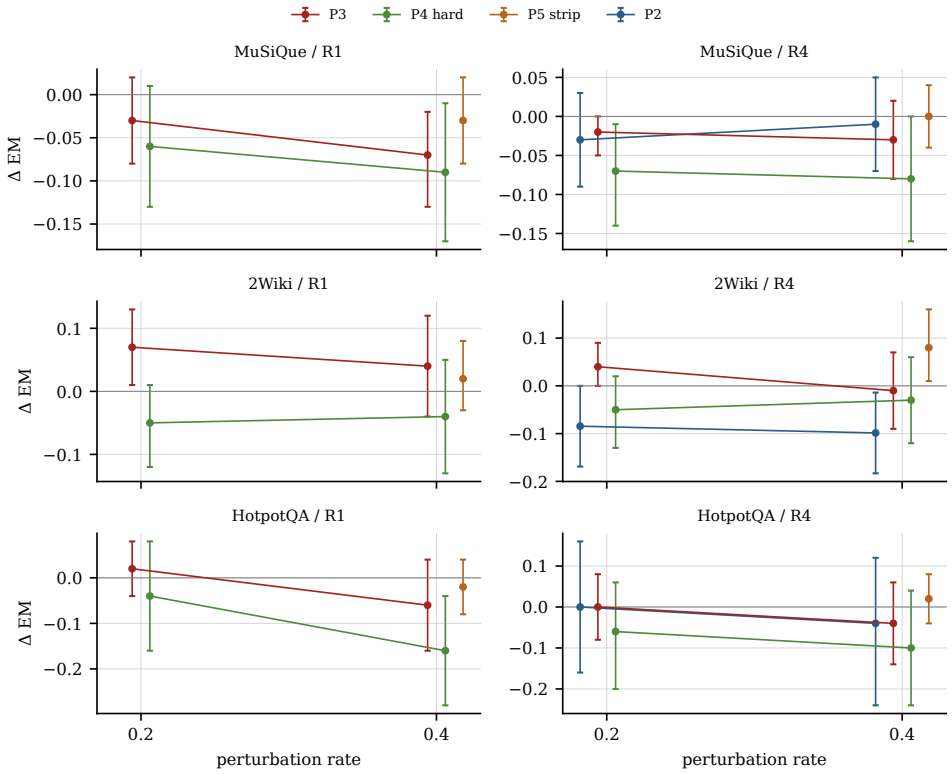


Figure 2: Exploratory pilot: change in exact match under label-preserving perturbations, relative to the same question unperturbed (95% CIs). P1 and P6 can remove the answer and are reported in Table 11 only.

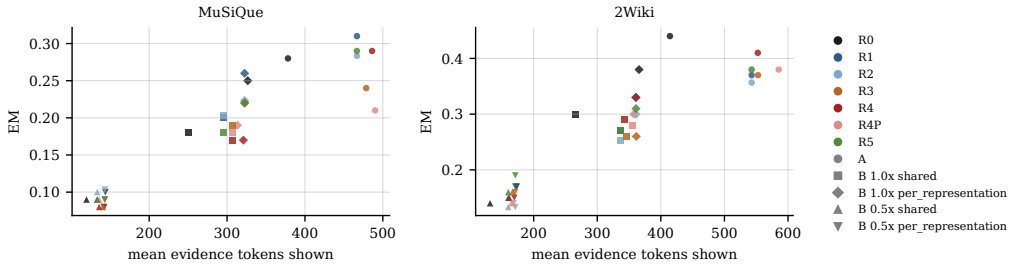


Figure 3: Exploratory pilot: exact match against the mean number of evidence tokens shown, for Protocol A (natural length) and Protocol B (budget $1.0\times$ or $0.5\times$ the content tokens; shared or per-representation policy).

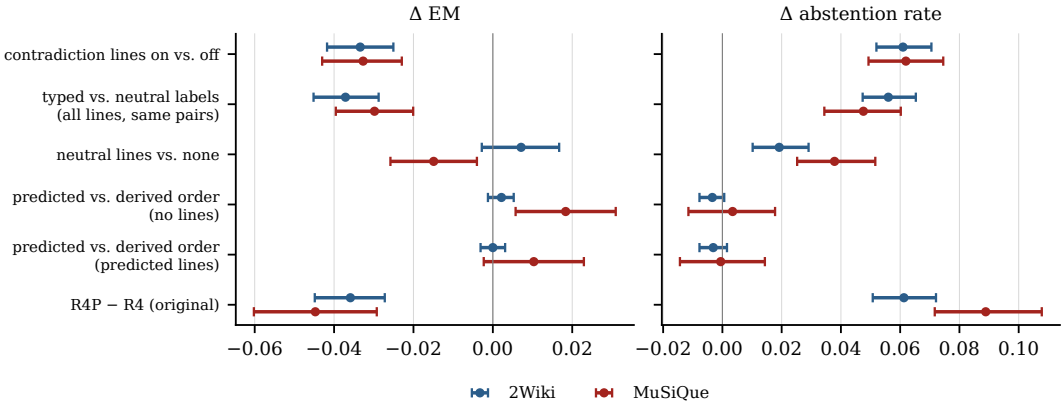


Figure 4: Fixed-order relation controls on the confirmatory questions (exploratory). Each row changes one factor on identical facts; points are paired differences with 95% CIs.

Data	Contradiction lines	n	ΔEM	95% CI	ΔAbst.	95% CI
2Wiki	0	1198	+0.000	[+0.000, +0.000]	+0.000	[+0.000, +0.000]
2Wiki	1-2	1015	-0.069	[-0.089, -0.049]	+0.088	[+0.068, +0.108]
2Wiki	3-5	577	-0.050	[-0.075, -0.024]	+0.137	[+0.106, +0.166]
2Wiki	6+	442	-0.020	[-0.043, +0.002]	+0.066	[+0.034, +0.097]
MuSiQue	0	734	+0.000	[+0.000, +0.000]	+0.000	[+0.000, +0.000]
MuSiQue	1-2	660	-0.053	[-0.073, -0.033]	+0.094	[+0.071, +0.118]
MuSiQue	3-5	257	-0.039	[-0.066, -0.016]	+0.132	[+0.089, +0.179]
MuSiQue	6+	93	-0.129	[-0.194, -0.065]	+0.129	[+0.065, +0.204]

Table 13: Effect of showing the predicted `CONTRADICTS` lines (R4P_FO−R4P_FO_NC) by the number of such lines in the prompt (exploratory). With no contradiction lines the two prompts are identical. Source: `revision/controls/relation_control_strata.csv`.

Data	Contrast	Qwen ΔEM	95% CI	ΔAbst.	Mistral ΔEM	95% CI	ΔAbst.	ΔLenient	95% CI
2Wiki	sentence items − passages (R1−R0)	-0.017	[-0.029, -0.005]	+0.052	-0.001	[-0.013, +0.010]	-0.001	-0.017	[-0.029, -0.005]
2Wiki	derived order (R5−R1)	-0.008	[-0.018, +0.002]	+0.012	+0.006	[-0.002, +0.015]	+0.008	+0.004	[-0.005, +0.013]
2Wiki	derived relation lines (R4−R5)	+0.006	[-0.002, +0.015]	+0.011	+0.008	[+0.000, +0.017]	-0.001	+0.015	[+0.006, +0.024]
2Wiki	predicted − derived lines, fixed order	-0.036	[-0.044, -0.027]	+0.064	-0.041	[-0.051, -0.031]	+0.063	-0.061	[-0.072, -0.051]
2Wiki	contradiction lines on − off	-0.033	[-0.041, -0.025]	+0.061	-0.034	[-0.043, -0.025]	+0.051	-0.050	[-0.061, -0.040]
MuSiQue	sentence items − passages (R1−R0)	-0.001	[-0.014, +0.014]	-0.009	+0.022	[+0.012, +0.033]	-0.046	+0.028	[+0.013, +0.042]
MuSiQue	derived order (R5−R1)	-0.011	[-0.026, +0.003]	-0.003	-0.002	[-0.013, +0.008]	+0.003	+0.003	[-0.011, +0.017]
MuSiQue	derived relation lines (R4−R5)	+0.010	[-0.003, +0.024]	-0.004	+0.003	[-0.007, +0.013]	-0.022	+0.018	[+0.002, +0.033]
MuSiQue	predicted − derived lines, fixed order	-0.055	[-0.068, -0.044]	+0.089	-0.025	[-0.035, -0.015]	+0.097	-0.057	[-0.073, -0.041]
MuSiQue	contradiction lines on − off	-0.033	[-0.042, -0.024]	+0.062	-0.015	[-0.023, -0.007]	+0.061	-0.033	[-0.045, -0.021]

Table 14: Second reader (exploratory cross-model replication on the confirmatory questions; n=3,232 2Wiki, 1,744 MuSiQue): Qwen2.5-7B-Instruct and Mistral-7B-Instruct-v0.3 with the same prompt and evidence. Lenient: gold alias contained in the answer (pre-specified before the Mistral run, because Mistral often answers in full sentences). Mistral levels: 2Wiki R0: EM 0.31, lenient 0.48, abstained 0.10; MuSiQue R0: EM 0.09, lenient 0.28, abstained 0.47. Source: `revision2/r2/reader2.csv`.

	`qa-v2` (frozen)	conservative	no abstention option
R0	0.395 / 0.202 / 0.403	0.330 / 0.307 / 0.363	0.473 / 0.006 / 0.521
R1	0.372 / 0.261 / 0.367	0.298 / 0.395 / 0.307	0.469 / 0.008 / 0.523
R4	0.378 / 0.274 / 0.348	0.314 / 0.422 / 0.264	0.487 / 0.006 / 0.507
R4P	0.342 / 0.328 / 0.330	0.281 / 0.473 / 0.246	0.470 / 0.009 / 0.521
R1−R0	-0.023 [-0.044, -0.003]	-0.032 [-0.052, -0.012]	-0.004 [-0.026, +0.017]
R4P−R4	-0.036 [-0.052, -0.019]	-0.033 [-0.047, -0.019]	-0.017 [-0.034, +0.000]
interaction, R1−R0	(qa-v2 − variant)	+0.009 [-0.014, +0.030]	-0.019 [-0.046, +0.007]
interaction, R4P−R4	(qa-v2 − variant)	-0.003 [-0.020, +0.014]	-0.019 [-0.039, +0.002]

Table 15: Abstention policy (exploratory; the same 1,000 confirmatory 2Wiki questions and evidence). Top: EM / abstention / wrong-answer rate per representation. Middle: within-prompt contrasts (ΔEM, 95% CI). Bottom: prompt × representation interaction, (contrast under `qa-v2`) − (contrast under the variant). Conservative: the last line replaced by *If the evidence does not fully establish the answer, reply "Answer: insufficient evidence". Do not guess.* Source: `revision2/r2/prompt3.csv`.

Data	Contrast	Ident.	ΔEM	95% CI	ΔAbst.
MuSiQue	R1−R0	0	+0.014	[+0.000, +0.028]	-0.006
MuSiQue	R5−R1	451	+0.010	[+0.000, +0.020]	-0.008
MuSiQue	R4−R5	342	+0.002	[-0.012, +0.016]	-0.002
MuSiQue	R4P−R4	72	-0.036	[-0.056, -0.020]	+0.068
MuSiQue	contradiction lines on−off	131	-0.022	[-0.038, -0.008]	+0.040
MuSiQue	R4P_FO−R5	86	-0.022	[-0.038, -0.008]	+0.052
2Wiki	R1−R0	5	-0.026	[-0.054, +0.002]	+0.072
2Wiki	R5−R1	498	+0.000	[+0.000, +0.000]	+0.000
2Wiki	R4−R5	322	+0.010	[-0.008, +0.028]	-0.006
2Wiki	R4P−R4	84	-0.030	[-0.054, -0.008]	+0.066
2Wiki	contradiction lines on−off	110	-0.024	[-0.046, -0.004]	+0.068
2Wiki	R4P_FO−R5	66	-0.016	[-0.042, +0.010]	+0.052

Table 16: Retrieved evidence, Protocol A (exploratory; fresh disjoint sample, n=500 per dataset; all representations render the same four reranked passages). Ident.: byte-identical prompt pairs. Full table with Protocol B in Appendix G. Source: `revision/retrieval/retrieval_contrasts.csv`.

Data	Rep.	Input tok.	Content tok.	Overhead tok.	Facts	Rel. lines	Titles shown	=R1	EM	Abst.
2Wiki	R0	560.7	388.2	29.3	12.46	0.0	2.5	44	0.408	0.189
2Wiki	R1	689.8	388.2	158.4	12.46	0.0	12.46	3232	0.391	0.241
2Wiki	R5	689.8	388.2	158.4	12.46	0.0	12.46	1858	0.383	0.253
2Wiki	R4	700.3	388.2	168.2	12.46	0.97	12.46	879	0.389	0.264
2Wiki	R4P	733.4	388.2	201.2	12.46	4.35	12.46	530	0.353	0.326
2Wiki	R4P_FO	733.5	388.2	201.2	12.46	4.35	12.46	510	0.353	0.329
2Wiki	R4P_FO_NC	703.4	388.2	171.3	12.46	1.37	12.46	851	0.386	0.268
2Wiki	R4N_FO	724.5	388.2	192.2	12.46	4.21	12.46	510	0.390	0.273
MuSiQue	R0	549.0	369.1	31.0	10.7	0.0	2.82	3	0.286	0.470
MuSiQue	R1	638.6	369.1	120.6	10.7	0.0	10.7	1744	0.285	0.461
MuSiQue	R5	638.6	369.1	120.6	10.7	0.0	10.7	522	0.274	0.458
MuSiQue	R4	657.6	369.1	138.6	10.7	1.82	10.7	0	0.284	0.454
MuSiQue	R4P	661.6	369.1	142.9	10.7	2.32	10.7	410	0.239	0.542
MuSiQue	R4P_FO	661.7	369.1	143.0	10.7	2.32	10.7	155	0.229	0.543
MuSiQue	R4P_FO_NC	646.4	369.1	127.9	10.7	0.78	10.7	307	0.262	0.481
MuSiQue	R4N_FO	657.4	369.1	138.7	10.7	2.3	10.7	155	0.259	0.495

Table 17: Serialisation accounting on the confirmatory questions (Protocol A, means per question). Content tokens are identical across representations by construction; overhead is identifiers, titles, headers and relation lines. Titles shown: source titles per prompt (once per passage in R0, once per item otherwise). =R1: questions whose evidence block is byte-identical to R1. Source: `revision2/serialization/serialization_accounting.csv`.

Data	Edge stratum	Contrast	n	Ident.	ΔEM	95% CI	ΔAbst.
2Wiki	exact anchor	R4−R5	1685	0	+0.008	[-0.005, +0.021]	+0.017
2Wiki	exact anchor	R5−R1	1685	693	-0.013	[-0.029, +0.003]	+0.015
2Wiki	fallback anchor	R4−R5	668	0	+0.007	[-0.015, +0.031]	+0.009
2Wiki	fallback anchor	R5−R1	668	286	-0.006	[-0.033, +0.021]	+0.021
MuSiQue	exact anchor	R4−R5	1224	0	+0.007	[-0.007, +0.023]	+0.002
MuSiQue	exact anchor	R5−R1	1224	386	+0.001	[-0.015, +0.016]	-0.017
MuSiQue	fallback anchor	R4−R5	520	0	+0.017	[-0.006, +0.042]	-0.017
MuSiQue	fallback anchor	R5−R1	520	136	-0.040	[-0.067, -0.012]	+0.029

Table 18: Derived structure by anchor quality (confirmatory questions; exploratory stratification by an input property). Exact anchor: every derived dependency is anchored to a sentence by exact string match; fallback: at least one endpoint is a fallback guess. Questions without derived edges (identical prompts) are omitted. Source: `revision2/serialization/anchor_strata.csv`.

Automatic triage (heuristic, not a judgement)	Pairs	Share
different source titles (different entities)	3,857	40%
same title, different years	1,962	20%
same title, different numbers	671	7%
same title, pronoun-initial sentence	734	8%
same title, other	2,526	26%
all predicted contradiction pairs	9,750	100%

Table 19: Predicted `CONTRADICTS` pairs on the confirmatory sample (R4P; 12,816 directed edges, 12,372 rendered, 3,044 questions). NLI confidence $\geq$0.90 (median 0.985). Categories are heuristic triage for prioritising human verification. Source: `revision/contradictions/contradictions_summary.json`.

Data	Prot.	Contrast	n	Ident.	ΔEM	95% CI	ΔAbst.	Tokens
MuSiQue	A	R1−R0	500	0	+0.014	[+0.000, +0.028]	-0.006	526→653
MuSiQue	A	R5−R1	500	451	+0.010	[+0.000, +0.020]	-0.008	653→653
MuSiQue	A	R4−R5	500	342	+0.002	[-0.012, +0.016]	-0.002	653→657
MuSiQue	A	R4P−R4	500	72	-0.036	[-0.056, -0.020]	+0.068	657→701
MuSiQue	A	contradiction lines on−off	500	131	-0.022	[-0.038, -0.008]	+0.040	665→701
MuSiQue	A	R4P_FO−R5	500	86	-0.022	[-0.038, -0.008]	+0.052	653→701
MuSiQue	B 1.0×	R1−R0	500	3	+0.022	[+0.002, +0.042]	+0.014	462→457
MuSiQue	B 1.0×	R5−R1	500	459	+0.002	[-0.004, +0.008]	-0.002	457→457
MuSiQue	B 1.0×	R4−R5	500	370	+0.006	[-0.004, +0.018]	-0.006	457→457
MuSiQue	B 1.0×	R4P−R4	500	133	-0.032	[-0.050, -0.016]	+0.052	457→452
MuSiQue	B 1.0×	contradiction lines on−off	500	201	-0.016	[-0.028, -0.006]	+0.026	455→452
MuSiQue	B 1.0×	R4P_FO−R5	500	144	-0.020	[-0.038, -0.004]	+0.038	457→452
2Wiki	A	R1−R0	500	5	-0.026	[-0.054, +0.002]	+0.072	525→681
2Wiki	A	R5−R1	500	498	+0.000	[+0.000, +0.000]	+0.000	681→681
2Wiki	A	R4−R5	500	322	+0.010	[-0.008, +0.028]	-0.006	681→685
2Wiki	A	R4P−R4	500	84	-0.030	[-0.054, -0.008]	+0.066	685→749
2Wiki	A	contradiction lines on−off	500	110	-0.024	[-0.046, -0.004]	+0.068	695→749
2Wiki	A	R4P_FO−R5	500	66	-0.016	[-0.042, +0.010]	+0.052	681→749
2Wiki	B 1.0×	R1−R0	500	23	+0.010	[-0.016, +0.034]	+0.030	454→451
2Wiki	B 1.0×	R5−R1	500	499	+0.000	[+0.000, +0.000]	+0.000	451→451
2Wiki	B 1.0×	R4−R5	500	383	-0.002	[-0.014, +0.010]	+0.004	451→451
2Wiki	B 1.0×	R4P−R4	500	157	-0.040	[-0.060, -0.020]	+0.054	451→447
2Wiki	B 1.0×	contradiction lines on−off	500	195	-0.036	[-0.054, -0.018]	+0.044	451→447
2Wiki	B 1.0×	R4P_FO−R5	500	141	-0.044	[-0.066, -0.022]	+0.064	451→447

Table 20: Retrieved evidence (exploratory, fresh disjoint sample of 500 questions per dataset): every representation renders the same four reranked passages; R4/R5 use annotation-derived edges restricted to retrieved sentence pairs (oracle structure on real retrieval). Protocol B: budget 1.0× content tokens, per-representation policy. Source: `revision/retrieval/retrieval_contrasts.csv`.

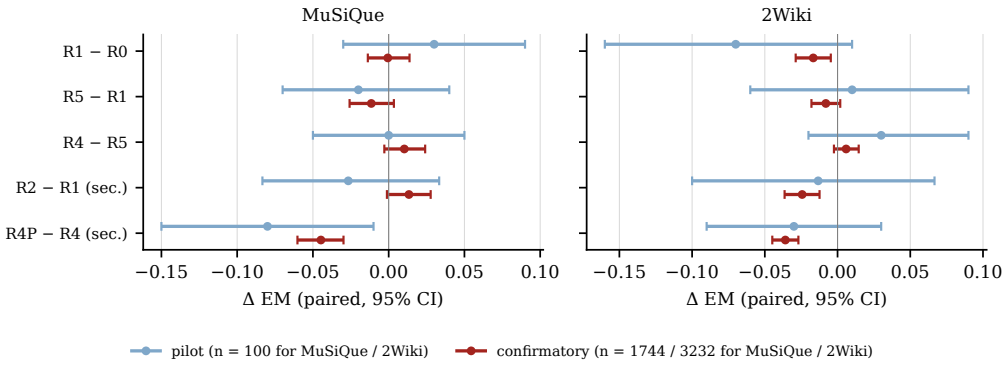


Figure 5: Protocol A contrasts in the exploratory pilot (n=100 per dataset) and on the disjoint confirmatory sample. The pilot estimate of R1−R0 on 2WikiMultiHopQA (-0.07) overstated the confirmed effect (-0.017) by a factor of four.

Data	Contrast	Stratum	n	ΔEM	95% CI	ΔAbst.
MuSiQue	R1−R0	retrieval: complete	149	+0.034	[-0.007, +0.074]	-0.034
MuSiQue	R1−R0	retrieval: partial	311	+0.006	[-0.006, +0.019]	+0.010
MuSiQue	R1−R0	retrieval: none	40	+0.000	[+0.000, +0.000]	-0.025
MuSiQue	R1−R0	distractor heavy: True	328	+0.006	[+0.000, +0.015]	+0.012
MuSiQue	R1−R0	long evidence: True	249	+0.012	[-0.008, +0.032]	-0.020
MuSiQue	contradiction lines on−off	retrieval: complete	149	-0.047	[-0.094, -0.007]	+0.047
MuSiQue	contradiction lines on−off	retrieval: partial	311	-0.013	[-0.026, -0.003]	+0.039
MuSiQue	contradiction lines on−off	retrieval: none	40	+0.000	[+0.000, +0.000]	+0.025
MuSiQue	contradiction lines on−off	distractor heavy: True	328	-0.012	[-0.024, -0.003]	+0.034
MuSiQue	contradiction lines on−off	long evidence: True	249	-0.016	[-0.040, +0.004]	+0.036
2Wiki	R1−R0	retrieval: complete	262	-0.057	[-0.099, -0.015]	+0.080
2Wiki	R1−R0	retrieval: partial	238	+0.008	[-0.025, +0.042]	+0.063
2Wiki	R1−R0	distractor heavy: True	119	+0.008	[+0.000, +0.025]	+0.067
2Wiki	R1−R0	long evidence: True	250	-0.036	[-0.072, +0.000]	+0.060
2Wiki	contradiction lines on−off	retrieval: complete	262	-0.019	[-0.050, +0.011]	+0.069
2Wiki	contradiction lines on−off	retrieval: partial	238	-0.029	[-0.055, -0.008]	+0.067
2Wiki	contradiction lines on−off	distractor heavy: True	119	-0.008	[-0.025, +0.000]	+0.050
2Wiki	contradiction lines on−off	long evidence: True	250	-0.012	[-0.040, +0.016]	+0.072

Table 21: Retrieved evidence, Protocol A, by retrospective retrieval outcome (gold labels used only to stratify). Exploratory; small strata have wide intervals. Source: `revision/retrieval/retrieval_contrasts.csv`.

Data	Sys.	EM	Cov.	Tok.	Units	Gold-pass. rec.	Ans. in ev.	Extr. calls	Latency (s)	CONTRA share
MuSiQue	B0	0.06	0.23	493	4.0	0.44	0.34	0.0	0.4	–
MuSiQue	B1	0.12	0.33	538	4.0	0.53	0.44	0.0	0.4	–
MuSiQue	B2	0.13	0.36	1390	57.1	0.53	0.44	15.3	24.3	–
MuSiQue	B3	0.15	0.38	1667	57.1	0.53	0.44	15.3	24.4	–
MuSiQue	B4	0.03	0.18	442	11.9	0.55	0.31	15.1	23.8	0.55
2Wiki	B0	0.23	0.54	506	4.0	0.76	0.69	0.0	0.4	–
2Wiki	B1	0.26	0.54	501	4.0	0.80	0.78	0.0	0.4	–
2Wiki	B2	0.24	0.48	1436	56.4	0.80	0.78	14.7	23.9	–
2Wiki	B3	0.20	0.50	1711	56.4	0.80	0.78	14.7	23.9	–
2Wiki	B4	0.11	0.25	489	11.9	0.80	0.56	14.7	23.8	0.62

Table 22: Experiment 0 with budgets and costs made explicit (pilot, n=100 per dataset). Budgets are *not* matched: B2/B3 show about three times the tokens of B0/B1; B4 selects at most 12 claims within 450 tokens. Latency includes LLM claim extraction (original-call latency for cached calls). CONTRA share: fraction of B4's predicted relations labelled `CONTRADICTS`. Source: `revision/analysis/C_exp0_costs.csv`.